Fiscal Year:	FY 2026	**Task Last Updated:**	FY 05/14/2026
PI Name:	Macias, Brandon Ph.D.		
Project Title:	Mitigating Headward Fluid Shifts with Veno-Constrictive Thigh Cuffs During Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281-483-2026
Organization Name:	NASA Johnson Space Center		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:	NOTE: Became civil servant fall 2020; previously KBR/NASA Johnson Space Center. Prior to that until 2016, was at the University of California, San Diego.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	05/15/2021	**End Date:**	05/14/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	Additions: Griffin Donley will replace the functional role of John Foulk. Katrina Carter, Jason Lytle, and Sarah Pickering, are new personnel part of the Cardiovascular and Vision Laboratory working on this project. Dr. Alex Huang from UCSD will join the team as Co-I to help with his OCT expertise and past thigh cuff countermeasure research expertise. Departures: John Foulk, Annelise Miller, Chris Miller, Evan Pettit, and Stuart MC Lee		
COI Name (Institution):	Laurie, Steven Ph.D. (KBR/NASA Johnson Space Center) Lee, Stuart Ph.D. (KBR/NASA Johnson Space Center) Cole, Chris Ph.D. (Clemson University) Huang, Alex (University of California, San Diego) Donley, Griffin (Clemson Textiles) Carter, Katrina (KBR/NASA Johnson Space Center) Lytle, Jason (KBR/NASA Johnson Space Center) Pickering, Sarah (KBR/NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This study will determine the ability of venoconstrictive thigh cuff (VTC) application for varying durations to mitigate the weightlessness-induced headward fluid shift during spaceflight for the purpose of evaluating this device as a countermeasure for SANS. The effects of varying durations of continuous VTC usage (~30 minutes, ~3 hours, and ~6 hours) on ocular and cardiovascular variables will be investigated. This proposed study will help determine the potential efficacy of VTC to mitigate the cephalad fluid shift that is hypothesized to be the primary initiating factor to SANS. The VTC countermeasure technique is low-mass, portable, simple to use and may be used for extended durations while working in an operational spaceflight environment. Specific Aim: 1) Determine the efficacy of VTC application to mitigate a spaceflight-induced headward fluid shift. We hypothesize that a VTC countermeasure will temporarily reverse or attenuate spaceflight-induced ocular and cardiovascular changes.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. This study will determine the ability of veno-constrictive thigh cuffs (VTC) application for varying durations to mitigate the weightlessness-induced headward fluid shift during spaceflight for the purpose of evaluating this device as a countermeasure for Spaceflight Associated Neuro-ocular Syndrome (SANS). The effects of varying durations of continuous VTC usage (30 minutes, 3 hours, and 6 hours) on ocular and cardiovascular variables will be investigated. This proposed study will help determine the potential efficacy of VTC to mitigate the cephalad fluid shift that is hypothesized to be the primary initiating factor to SANS. The VTC countermeasure technique is low-mass, portable, simple to use and may be used for extended durations while working in an operational spaceflight environment. The investigator team has the required expertise to conduct this research and has experience from a ground study led by this team in 33 subjects who wore these VTC for up to ~2 hours while supine at rest. An updated VTC and microfiber sleeve have been developed and tested in these ground-based studies.
Research Impact/Earth Benefits:	The Flight Thigh Cuff research study will advance NASA's understanding of use of the VTC as a potential countermeasure for Spaceflight Associated Neuro-ocular Syndrome (SANS), an important human health and performance risk. This goal will be accomplished by quantifying vascular and ocular changes before and during spaceflight. Given the unique environment of the International Space Station, commonly used ophthalmic instruments are being implemented in novel operational environments (e.g., optical coherence tomography (OCT) and pneumotonometry). This work will help NASA to evaluate this spaceflight-associated risk more accurately and to refine its data-driven approach to SANS countermeasure development.
Task Progress:	INTRODUCTION Structural and functional changes of the eye that develop in many astronauts during extended periods of weightlessness are described collectively as spaceflight-associated neuro-ocular syndrome (SANS). Venoconstrictive thigh cuffs (VTC) are a mechanical countermeasure capable of reducing headward fluid shifts in both ground-based and spaceflight studies, and thus, may be a countermeasure against the development of SANS. PURPOSE The purpose of this study is to determine the efficacy of short duration VTC application to mitigate a spaceflight-induced headward fluid shift. We hypothesize that a VTC countermeasure will temporarily reverse or attenuate spaceflight-induced ocular and cardiovascular changes. In addition, we will investigate use of VTC for up to six hours of duration during spaceflight to characterize the role duration of use has on our outcome variables. METHODS This study will evaluate the effectiveness of an acute VTC countermeasure application (hours) to affect ocular and cardiovascular variables, including intraocular pressure (IOP), ocular structure, and vascular fluid shifts, using state-of-the-art techniques. Eight astronauts were recruited to participate in this flight study. Astronauts were studied before and during approximately 180-day International Space Station (ISS) spaceflight missions. Multiple sizes of VTC were manufactured, flight-certified, and returned to the NASA Johnson Space Center (JSC) Cardiovascular and Vision Laboratory (CVL) for baseline VTC fit check and data collection procedures. Before flight, crewmembers participated in a VTC fit check session approximately 270 days before launch. This session included bilateral leg circumference measurements, donning of VTC to determine the correct size of VTC, and use of Doppler ultrasound to assess femoral vein blood flow during use of VTC at pressures ranging from 50 mmHg down to 0 mmHg in 10 mmHg step increments. At each pressure, crewmembers will plantar flex their ankle to determine if muscle pump action can lead to a return of venous blood flow past the VTC. Preflight data collection occurred approximately 90 days before launch (Table 1) in the seated, supine, and supine with VTC postures. Outcome variables included optical coherence tomography (OCT) image analysis (total retinal thickness and choroid thickness), IOP, ultrasound (internal jugular vein area and pressure, stroke volume and cardiac output, femoral vein flow characteristics), brachial blood pressure, heart rate, and thigh circumference. The inflight study assessed these same variables before and after application of VTC in weightlessness. During flight, a VTC fit check session occurred at ~Flight Day (FD) 30 to reassess the appropriate VTC size during spaceflight. On ~FD45 and ~45 days before return (R-45), baseline measures without VTC were followed by application of VTC with data collections occurring at three timepoints (30 minutes, 3 hours, and 6 hours) to characterize the temporal profile of key fluid shift outcome measures. Inflight data collections with VTC were conducted early (~FD45) and late (~R-45) inflight to determine if mission duration affects VTC fit and efficacy of the fluid redistribution. If a mission extension of greater than 30 days occurred after the R-45 session had already occurred, the R-45 session was repeated and has a new window of 45 days before return. This project also will share medically relevant test results already collected by the Space Medicine Operations Division to complement the research measures. RESULTS All data collection has been completed on 8 participants. Data have been analyzed and are in preparation for a manuscript and publication. The original plan to study 10 participants was reduced to allow for the acceleration of a countermeasure arm to this study. For this countermeasure arm, a new protocol was developed to test the VTC countermeasure throughout a spaceflight mission. We have since updated consent form and pitched opportunity to 5 crew members. This new arm uses a SANS prediction tool to target those we anticipate will develop SANS. Two participants have consented to the screening procedure but were deemed unqualified to participate. MISSION IMPACT This study investigated if use of VTC for up to six hours can acutely mitigate the headward fluid shift that develops during exposure to weightlessness. The results will narrow the SANS403 (test candidate countermeasure(s) in a spaceflight environment) knowledge gap described in the Human Research Roadmap and, in turn, will help NASA to evaluate this spaceflight-associated risk more accurately and to refine its data-driven approach to SANS countermeasure development. The experience gained from this study also enabled the VTC to be supplied for medical use early in the mission as a countermeasure of space adaptation symptoms. While the outcome measures presented here are novel, use of similar VTC (Braslets) provided to crewmembers who fly on Soyuz vehicles

	demonstrates anecdotal evidence that a VTC countermeasure can be operationally utilized without significant risk. Presentation M.J. Poczatek, C.S. O'Grady, G.S. Boothe, S.S. Laurie, D.S. Martin, S.M.C. Lee, C. Cole, G. Donley, M.H. Young, B.R. Macias. Mitigating Headward Fluid Shifts with Veno-Constrictive Thigh Cuffs During Spaceflight (Flight Thigh Cuff Study). NASA HRP Investigators Workshop 2025. (Conference abstract)
Bibliography Type:	Description: (Last Updated: 06/03/2026)
Abstracts for Journals and Proceedings	Poczatek MJ, O'Grady CS, Boothe GS, Laurie SS, Martin DS, Lee SMC, Cole C, Donley G, Young MH, Macias BR. "Mitigating headward fluid shifts with veno-constrictive thigh cuffs during spaceflight (flight thigh cuff study)." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025